

Deliverable 1: Initial Planning and Design Report

Cross-Language Application Development

Project Title

Expense Tracker Application

Programming Languages

Python and C++

1. Application Design

1.1 Overview

The project selected for this assignment is an Expense Tracker Application. The application will be written in Python as well as C++ with the same functionalities. Creating the same program in two programming languages will give a chance to compare syntax, programming paradigm, performance, readability, maintainability and language-specific features. Expenses Tracker Application will assist the users in efficiently logging expenses and organizing financial data. Users will be able to add new expenses to the system, edit any existing records, delete unwanted records, search for expenses, filter expenses by category or date, and be able to view a summary report of expenses which details total expenses. Both versions will give the same results, and use the good aspects of the programming language.

1.2 System Architecture

The application will follow a three-layer architecture consisting of:

User Interface

The UI will offer a simple menu driven system with which the user can interact with the application. Users will choose from options to add expenses, search records, create reports, and exit the program. The interface will be user-friendly and intuitive.

Business Logic

All user requests will be processed in the Business Logic layer. It will validate user data, sum up expenses, filter expenses according to user needs, search for expenses and create reports. This layer helps to ensure that all business rules are applied properly before the data is stored or displayed.

Data Storage

The Data Storage layer will maintain all expense records. Each expense will contain:

- Expense ID
- Date
- Amount
- Category
- Description

The stored information will allow users to retrieve, update, and analyze their financial records efficiently.

1.3 Functional Requirements

The Expense Tracker Application will provide the following features:

- Add new expenses
- View all recorded expenses
- Search expenses by category
- Search expenses by date

- Update expense information
- Delete expense records
- Filter expenses using different criteria
- Calculate total expenses
- Generate category-based summary reports
- Validate user input
- Handle invalid entries using exception handling

Both implementations will offer the same functionality so that it is a fair test between the two programming languages.

1.4 Language-Specific Features

Although both implementations will perform the same tasks, each language will demonstrate its own strengths. Dynamic typing, dictionaries, lists, and datetime library will be utilized for Python implementation, which will make date management easy and make the code more readable. Python's short syntax will enable fast development without having to write a lot of code. The C++ version will exemplify OOP concepts, robust statically-typed programming, classes, vector containers and the STL (Standard Template Library). C++ will provide some memory management and data structures that will be used to enhance the performance of the applications.

2. GitHub Repository

A GitHub repository will be created to manage the development process. The repository will include documentation, screenshots, reports, resources for the project, along with different folders for Python and C++ implementations. Individuals from each team will work individually on their respective feature branch and then complete them with pull requests back to the main

branch. Project progress will be tracked, tasks assigned, bugs will be recorded and the project milestones will be managed using GitHub Issues and the Project Board. All changes will be documented throughout the development process with version control.

3. Task Assignment

The project work will be shared equally between both team members.

Member 1

- Design overall application architecture.
- Develop the Python implementation.
- Prepare project documentation.
- Review C++ implementation.

Member 2

- Develop the C++ implementation.
- Perform testing and debugging.
- Compare both implementations.
- Prepare presentation materials.

Both members will work together to define project requirements, review code, test the application, resolve identified issues, and complete the final report and presentation.

4. Timeline

Day 1 - Planning and Design

- Review assignment requirements.
- Design system architecture.
- Create GitHub repository.
- Assign responsibilities.

- Design application data structures.
- Develop the application menu.

Day 2 - Core Development

- Implement expense management functions.
- Implement searching and filtering.
- Generate summary reports.
- Validate user input.
- Conduct initial testing.

Day 3 - Testing and Documentation

- Complete debugging.
- Perform comprehensive testing.
- Compare both language implementations.
- Capture application screenshots.
- Finalize documentation.
- Prepare the presentation.

5. Documentation of Design

The documentation of all design decisions will be carried out during the project. The application architecture, user interface, how data is stored, business logic, data validation, testing, and implementation choices will be included in the documentation. A comparison of the implementation in Python and C++ will be included in the special attention. The syntax, programming paradigms, readability and maintainability of the code, memory management, performance, scalability, exception handling and use of language-specific features will be compared. Difficulties to be faced in the process of development and lessons learnt from every

development milestone will also be recorded. Keeping detailed records during the development process will help to maintain consistency of both implementations and will give enough evidence for the final comparison report.